

Final report - Sprint 13 | Data Analysis Bootcamp

Abstract

This project analyzes how e-commerce delivery options can be compared using route, emissions, logistics tracking and Barcelona urban traffic data. The goal is not to build a real route optimization system, but to demonstrate a complete data analysis workflow: loading, cleaning, transformation, EDA, KPI calculation, visualization, scoring and basic validation. The main output is a transparent educational Delivery Option Score. The best overall option is Route option 418, with a score of 92.62, time of 15.33 minutes, cost of 85.07, estimated CO2 of 2.04 kgCO2e and reliability of 0.863.

1. Introduction

The practical question behind this project is: when an e-commerce company has several delivery options, should it always choose the fastest or cheapest one? In urban logistics, the answer is not so simple. A route can be fast but expensive, cheap but less reliable, or operationally efficient but more polluting.

For that reason, this project analyzes route options from a multi-criteria perspective. Kamport is only a possible applied context; the focus of the report is the data analysis process and how data can support more transparent delivery decisions.

2. Methodology

The work follows an IMRaD logic adapted to a bootcamp final project: problem, data, results and interpretation. The analysis was developed in Jupyter Notebook using Python, pandas, NumPy, Matplotlib, Seaborn, Plotly and SciPy.

Dataset	Source	Format	Role in the project
E-commerce Logistics Route Planning	Kaggle	CSV	1,000 routes: time, cost, traffic and reliability
Green Logistics Carbon Footprint	Kaggle	CSV	12,000 deliveries: CO2, vehicle and traffic
Transportation and Logistics Tracking	Kaggle	XLSX	6,877 cleaned deliveries: ETA, delay and on-time
Barcelona TRAMS Traffic	Open Data BCN	CSV	6.15M valid records: urban traffic context

3. Main results

Main project KPIs

Visual summary of the four analysis dimensions.

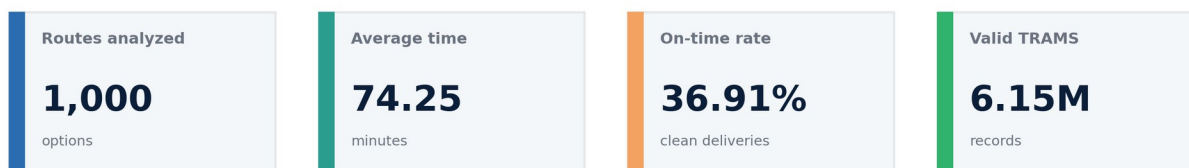


Figure 1. Main project KPIs. Source: own analysis based on notebook outputs.

3.1 Cost-time-score trade-off

The first important result is that the best routes are not explained by one variable only. In general, higher-scoring options appear in the lower-time and lower-cost area, but reliability, traffic and CO2 also influence the final position.

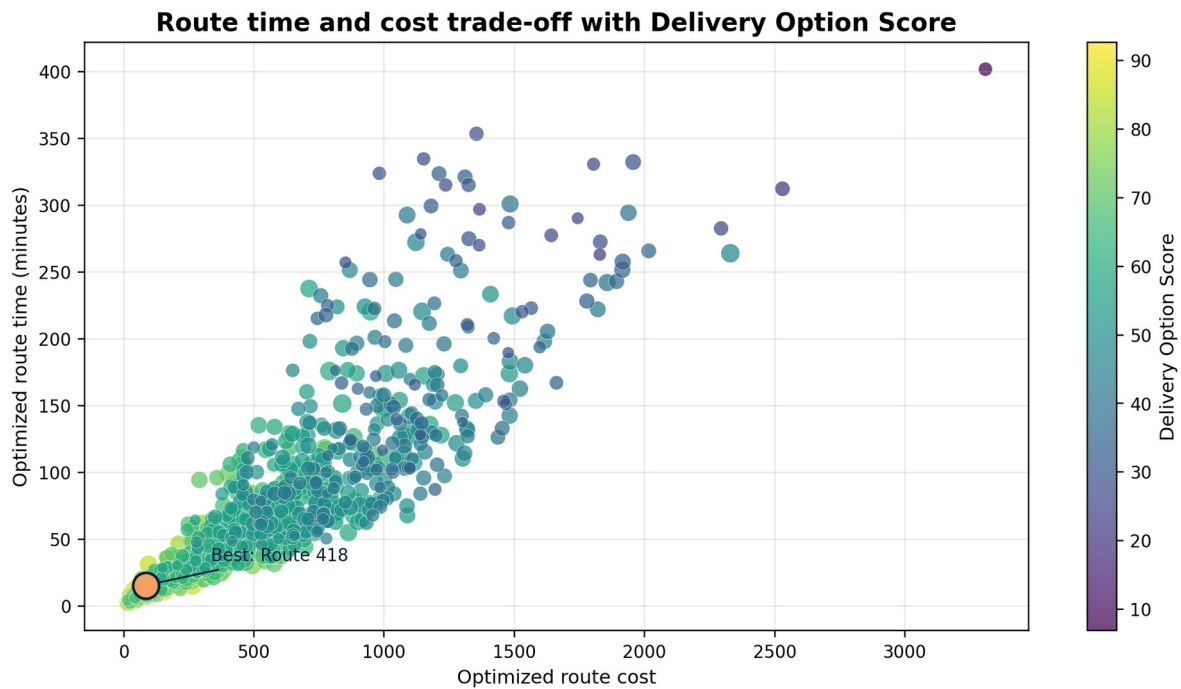


Figure 2. Route cost-time trade-off with Delivery Option Score. Lighter colors indicate higher scores; point size reflects reliability.

3.2 Delivery Option Score

The Delivery Option Score combines six normalized components: time (20%), cost (20%), estimated CO2 (25%), reliability (20%), traffic (10%) and efficiency (5%). It is an educational and explainable model; the weights are project assumptions, not industry standards.

Best route	Score	Time	Cost	Estimated CO2	Reliability
Route option 418	92.62	15.33 min	85.07	2.04 kgCO2e	0.863

4. Score, categories and decision matrix

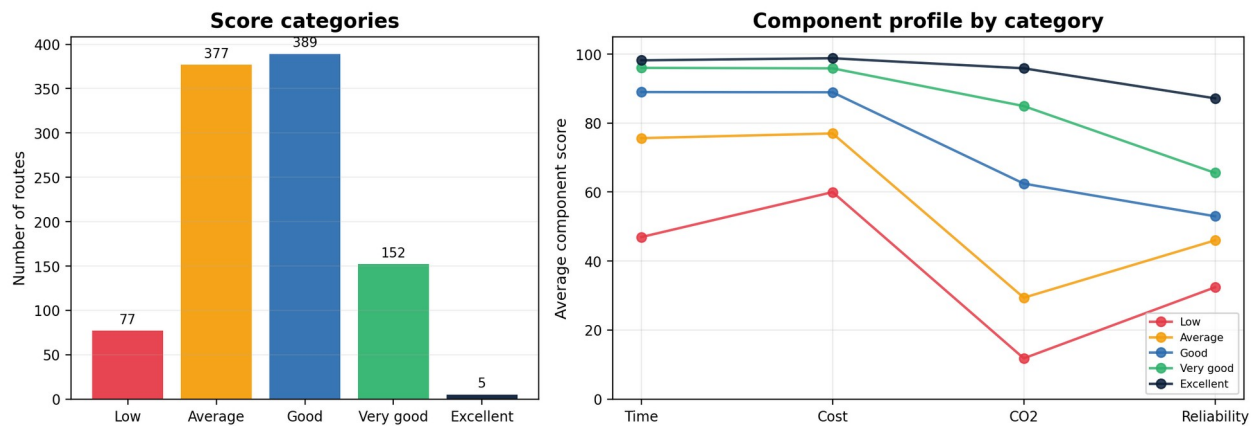


Figure 3. Score distribution and component profile by category. Only 5 routes reach Excellent, which shows that the score is selective.

The decision matrix helps explain that the best route depends on the decision need. Route option 418 is the best overall option, Route option 715 stands out for speed and efficiency, and Route option 350 is the best for cost.

Need	Route	Score	Time	Cost	CO2	Reliability
Overall balance	418	92.62	15.33	85.07	2.04	0.863
Speed	715	88.95	1.43	16.70	0.75	0.669
Cost	350	88.12	2.32	15.29	1.05	0.729
Low traffic	80	56.52	153.95	1059.67	27.18	0.742
Reliability	158	84.48	22.26	143.37	12.63	0.896
Efficiency	715	88.95	1.43	16.70	0.75	0.669

5. CO2, vehicles and green fleet scenario

The sustainability analysis shows that vehicle type strongly changes emission intensity. Heavy Truck and Diesel Vans have the highest intensities, while Cargo Bicycle, Drone Delivery and Electric Van show much lower values.

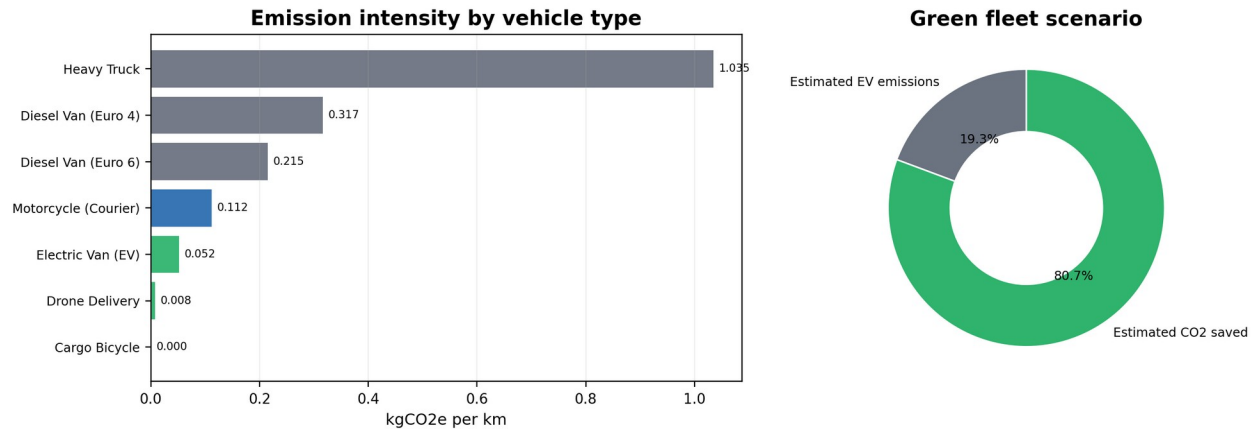


Figure 4. Emission intensity by vehicle and green fleet scenario. Source: `emission_intensity_by_vehicle.csv` and `green_fleet_transition_scenario.csv`.

Exploratory green fleet scenario

If 50% of Diesel Van trips were replaced by Electric Van trips, the scenario estimates 30,167.33 kgCO2e saved (80.67%) for the replaced trips. This is an educational estimate, not a complete operational simulation.

The main reading is that environmental improvement does not only depend on reducing kilometers: it also depends on vehicle type and how routes are assigned.

6. Traffic, reliability and short validation

In the tracking dataset, the clean on-time delivery rate is 36.91%, while the clean delayed rate is 63.09%. In Barcelona TRAMS, after excluding No data, the most common status is Fluid (50.72%), followed by Very fluid (32.67%).

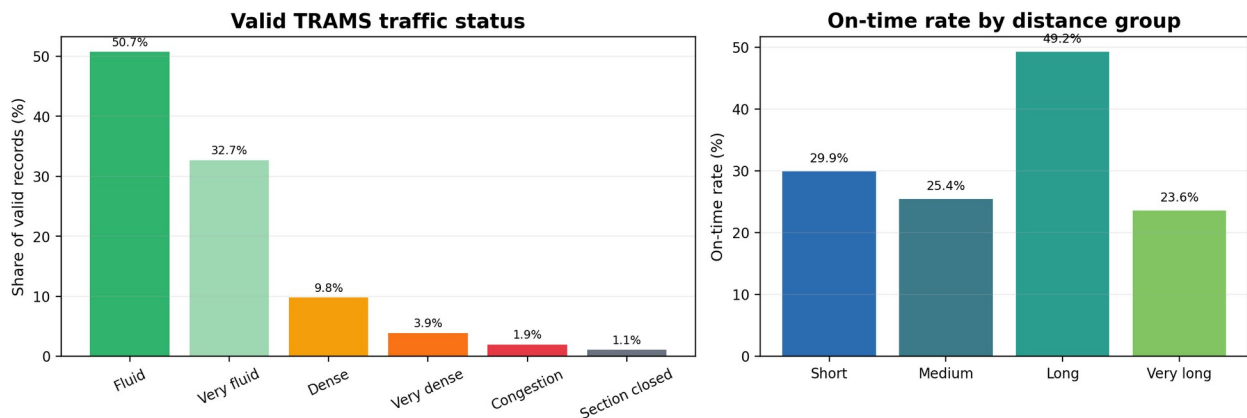


Figure 5. Valid traffic status and on-time rate by distance group. Source: `TRAMS` and `tracking outputs`.

As an additional check, Spearman correlations were calculated. The strongest relationship is distance vs CO2 ($\rho = 0.93$). The score has strong negative relationships with cost (-0.85), estimated CO2 (-0.84) and time (-0.83), which confirms that the indicator behaves consistently with its design.

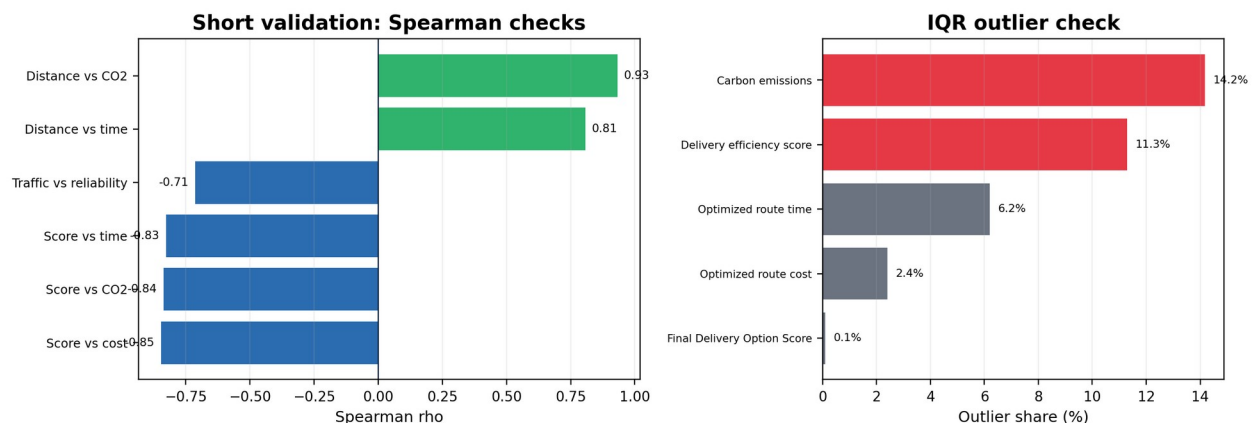


Figure 6. Short validation with Spearman and IQR outliers. Source: `statistical_relationship_checks.csv` and `iqr_outlier_summary.csv`.

7. Discussion

The results suggest that sustainable delivery decisions need to balance several criteria. Looking only at time can ignore cost, reliability or emissions. Looking only at cost can miss service quality. The score does not solve every decision, but it helps structure the problem transparently. The analysis also shows that distance is not enough: vehicle type, traffic and operational reliability also matter.

8. Limitations and future work

- The datasets come from different sources and were analyzed in modules, not through a row-by-row operational merge.
- The Delivery Option Score uses project-defined weights and should be interpreted as an educational model.
- Estimated CO2 uses an average factor per kilometer.
- TRAMS is used as urban context, not as direct route-level input.
- No real Kamport, carrier or live route data is used.
- Future work could add real order data, ITINERARIS, predictive models and an interactive dashboard.

9. Conclusion

This project demonstrates how a data analysis workflow can support urban logistics and e-commerce delivery decisions. The main conclusion is that a route should not be evaluated only by speed or cost. The Delivery Option Score combines time, cost, estimated CO2, traffic, reliability and efficiency in a simple and explainable way. It is not a production algorithm, but it helps show the value of data for analyzing logistics trade-offs.

Executive summary

The project compares e-commerce delivery routes using operational and sustainability indicators. The best overall option is Route option 418, with a score of 92.62. The analysis also shows a clear emission gap between vehicle types and estimates that replacing 50% of Diesel Van trips with Electric Van trips could save 30,167 kgCO2e in the simplified scenario.

Main figures and recommended appendix

Main figures: cost-time-score trade-off, score categories and components, decision matrix, emission intensity, green fleet scenario, traffic/reliability and short validation. Recommended GitHub appendix: top 10 routes, full decision matrix, IQR outliers, Kruskal-Wallis, monthly congestion curve and output CSV list.

Short defense outline

1. Practical question and context	6. CO2 and vehicle findings
2. Datasets used	7. Traffic and reliability
3. Analysis workflow	8. Short validation
4. Main KPIs	9. Limitations
5. Delivery Option Score	10. Conclusion and future work

Outputs used in the report

File	Main use
delivery_option_score_results.csv	Final score, ranking and best route
route_decision_matrix.csv	Comparison by decision need
emission_intensity_by_vehicle.csv	CO2 intensity by vehicle
green_fleet_transition_scenario.csv	50% Diesel Van to Electric Van scenario
barcelona_trams_valid_status_summary.csv	Urban traffic context
on_time_rate_by_distance_group.csv	Reliability by distance group
statistical_relationship_checks.csv	Spearman validation
iqr_outlier_summary.csv	Outlier check

References

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